



Decoding the drivers of urban road material metabolism and environmental impacts: an integrated MFA-LCA-LMDI analysis of Shanghai, China

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ABSTRACT

Urbanization has accelerated material accumulation in urban road systems, leading to increasing environmental pressures. Existing studies mainly focus on demand-side drivers, while rarely incorporating infrastructure design variables into material stock analysis. Using Shanghai's road system during 2014–2023 as a case study, this study develops an integrated MFA-LMDI-LCA framework to quantify road material stocks, decompose driving mechanisms, and assess cradle-to-construction environmental impacts. Results show that stocks increased from 416.75 Mt in 2014 to 810.61 Mt in 2023, with larger absolute stocks in outer districts but substantially higher area-normalized stock densities in central districts. Decomposition results indicate that road length effect is the dominant driver, while road width effect partially offsets material stock growth. Environmental impacts, represented by the normalized and weighted ReCiPe 2016(H) endpoint single score (points), increased with material accumulation, including a 20.69% year-on-year increase from 2015 to 2016. The framework clarifies the roles of scale-driven, design-related efficiency and socio-structural factors, and provides new insights for resource-efficient urban infrastructure planning.

1. Introduction

Urban road systems have become a major repository of construction materials in rapidly urbanizing regions, giving rise to substantial and long-lived in-use material stocks (Kloostera et al., 2022; Dai et al., 2022; Ye et al., 2026). Previous studies have shown that the construction and maintenance of road infrastructure require substantial quantities of asphalt and aggregates, which inevitably exert significant environmental burdens through resource extraction, processing, and downstream emissions (Zhao et al., 2023; Zhang et al., 2023). Once incorporated into road systems, these materials are locked into cities for decades, functioning as both critical mineral assets and persistent sources of environmental burden. As urban road networks continue to expand and mature, understanding the scale, structure, and environmental implications of road-related material stocks has become an essential prerequisite for advancing resource-efficient and low-carbon urban infrastructure systems. In large global cities with well-established road networks, such as New York, London, and Chicago, urban land expansion has remained relatively stable over recent decades (Zhao et al., 2017). Against this

background, the continued growth of road-related material stocks raises a fundamental yet unresolved question: to what extent is material accumulation governed by socio-structural factors associated with population dynamics and urban development stages; to what extent is it driven by scale-driven processes such as road network expansion and infrastructure provision; and to what extent can it be moderated by design-related efficiency mechanisms embedded in internal infrastructure design and material utilization?

This uncertainty has important implications for resource and environmental governance. Existing studies on urban material stocks and infrastructure metabolism have demonstrated that economic growth and population dynamics are key drivers of stock accumulation at national scales (Deng et al., 2022; Tareq et al., 2025). However, these analyses largely treat road systems as homogeneous assets and focus on external demand-side factors, providing limited insight into the role of internal road characteristics such as road width, cross-section design, and material intensity. Previous research suggests that design-related parameters can significantly influence infrastructure material stocks, as road width and pavement structure determine the geometric dimen-

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sions of road cross-sections and the material quantities required for each structural layer (Yu et al., 2021). Therefore, integrating these design-related factors into large-scale infrastructure stock analyses is essential for enhancing understanding of the factors shaping infrastructure accumulation.

Within the framework of urban material metabolism, Wolman (1965) described urban processes as metabolic functions, where a city acquires energy and materials from its external environment and produces associated waste emissions. Material Flow Analysis (MFA) has therefore emerged as a key methodological approach for quantifying material inputs, flows, and in-use stocks (Meijer et al., 2024). To operationalize this quantification, MFA can be implemented through either top-down or bottom-up approaches (Yang et al., 2023; Zhong et al., 2022). While top-down approaches rely on aggregated statistical data, they are often constrained in their ability to achieve detailed quantification required for urban-scale material stocks (Cai et al., 2024). In contrast, bottom-up approaches were adopted to enable more detailed assessments by integrating specific service units with material intensity data (Mollai et al., 2021; Grossegger et al., 2024). However, they lack the inherent capability to represent spatial distribution and geographic heterogeneity. Geographic Information Systems (GIS) addresses this limitation by enabling the precise mapping and spatial analysis of where these stocks are accumulated. Previous studies have applied GIS-based MFA to estimate material stocks of buildings and infrastructures across multiple spatial scales. For instance, Mao et al. (2020) produced high-resolution maps of urban built-environment stocks in Beijing, while Wiedenhofer et al. (2024a) quantified building and mobility infrastructure stocks at the national scale for the United Kingdom and the Republic of Ireland. Despite these advances, most existing MFA studies treat road infrastructure as part of aggregated built environment categories, offering limited insight into the internal structural characteristics of urban road system.

Beyond quantifying material stocks, understanding the mechanisms driving their growth is essential for assessing whether material accumulation can be moderated through policy or design interventions. Structural Decomposition Analysis (SDA) and Index Decomposition Analysis (IDA) have been widely employed to disentangle the contributions of socioeconomic and technological factors to changes in resource use and emissions. Among these methods, the Logarithmic Mean Divisia Index (LMDI) method has gained prominence due to its analytical robustness, data simplicity, and zero residual errors (Gu et al., 2019; Hasan and Liu, 2022). It has been widely used to identify influencing factors for carbon emissions (Ortega-Ruiz et al., 2020; Gao et al., 2024; Zhao et al., 2025) and energy consumption (Li et al., 2019; Chong et al., 2023). In the context of material stocks, LMDI-based studies have primarily focused on socio-structural and scale-driven drivers, such as population growth, economic development, and technological changes (Chu et al., 2022; Dai and Yue, 2023). Empirical applications include analyses of housing vacancy stock in Chinese Shandong Peninsula urban agglomeration (Yang et al., 2023), and material stock of urban infrastructures in China (Zhang et al., 2019). However, design-related efficiency mechanisms embedded in internal road design characteristics, such as road width and material utilization intensity, have rarely been incorporated in decomposition analyses.

The environmental consequences of road material stock growth have been widely examined using Life Cycle Assessment (LCA), which provides a systematic framework for quantifying environmental impacts across the life cycle of road infrastructure, including material production, transportation, construction, operation, and demolition stages (Qiao et al., 2022). Numerous studies have applied LCA to evaluate the potential environmental impacts associated with different pavement structures and materials (Aghel et al., 2025; De Pascale et al., 2025). Among the various impact indicators, carbon emissions are the most frequently examined, with assessments conducted under varying life-cycle scopes, ranging from specific stages to the entire life cycle (Yu et al., 2024; Xu et al., 2025; Huang et al., 2022). Beyond carbon-focused assessments, Hossain et al. (2022) assessed broader environmental impacts

including greenhouse gas emissions for different pavement types in Hong Kong. Despite these contributions, most existing LCA studies of road infrastructure focus predominantly on greenhouse gas emissions, while other environmental impact categories, such as eutrophication, acidification, and resource depletion, receive comparatively less attention. This focus limits the ability to fully assess whether efficiency-oriented design strategies that reduce material demand also translate into broader environmental benefits across multiple impact dimensions.

To overcome the limitations of single-method approaches, recent studies have increasingly integrated GIS, MFA, and LCA to assess material stocks and their associated environmental impacts in the built environment. Wiedenhofer et al. (2024b) applied the GIS-MFA-LCA framework to quantify global infrastructure material stocks and embodied carbon emissions, while Wang et al. (2023) assessed material stocks and greenhouse gas emissions reduction potential of asphalt and concrete roads in Belgium. Similar integrated approaches have been used to analyze buildings material stocks in Macao and China (Liu et al., 2023; Long et al., 2024). While these integrated studies represent important methodological advances, most focus on quantifying material stocks and environmental impacts without explicitly linking changes in material accumulation to their underlying driving mechanisms. Consequently, integrated frameworks capable of simultaneously quantifying road material stocks, identifying their driving factors, and evaluating their environmental consequences remain relatively limited.

To address this unresolved issue, this study develops an integrated analytical framework that combines MFA, LCA, and LMDI method to examine urban road material metabolism and its environmental consequences. This framework enables a quantitative distinction among scale-driven, design-related efficiency, and socio-structural effects. This study not only quantifies the spatiotemporal evolution of road material stocks and their life cycle environmental impacts across multiple impact categories, but also provides empirical evidence on the extent to which design-level efficiency gains can moderate material accumulation under sustained economic growth. The findings offer transferable insights into the limits and potentials of efficiency-oriented strategies for resource governance in mature urban road system.

2. Methodology

2.1. Model framework

This study proposes an integrated analytical framework (Fig. 1) that combines Geographic Information System (GIS)-based spatial processing, bottom-up Material Flow Analysis (MFA), Logarithmic Mean Divisia Index (LMDI) decomposition, and Life Cycle Assessment (LCA) to systematically examine urban road material metabolism. The framework is designed as a modular and transferable approach that can be applied to any city with available road network and socio-economic data. It links physical infrastructure geometry, material composition, economic drivers, and environmental impacts within a unified analytical structure.

The framework consists of four interrelated modules. (1) Spatial data processing module (GIS-based): Vector road network data and administrative boundary data were processed to derive road length by grade, road area, and intersection corrections. (2) Material stock estimation module (MFA-based): Using bottom-up geometric calculations and pavement structural parameters, material stocks were quantified for each road grade and material type. (3) Driving force decomposition module (LMDI-based): Changes in material stocks were decomposed into additive contributions from scale-driven, design-related efficiency, and socio-structural factors. (4) Environmental impact assessment module (LCA-based): Material stock outputs were translated into environmental burdens across cradle-to-construction stages. By integrating physical stock accounting with driver decomposition and environmental impact assessment, the framework enables a comprehensive examination of whether urban road material growth is primarily scale-driven or moderated by internal efficiency signals.

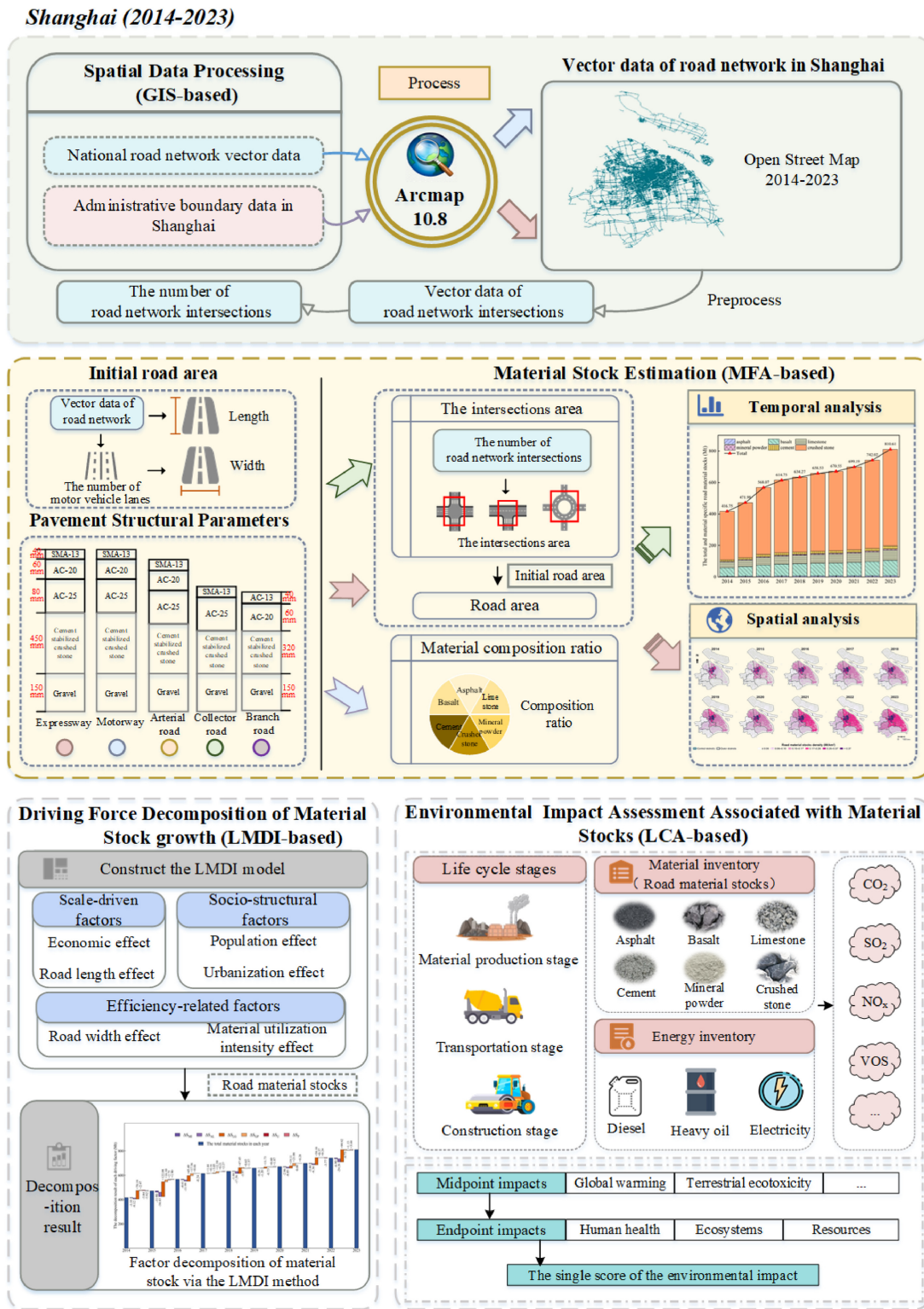


Fig. 1. Integrated MFA-LMDI-LCA research framework.

2.2. Road network data acquisition and preprocessing

Vector-based road network data containing geometric and attribute information (e.g., road names and classifications) were obtained from open-source platforms (e.g., OpenStreetMap), governmental geographic databases, or official spatial datasets. To ensure spatial consistency with the target urban system, the administrative boundary vector data were used to clip the road network data in Arcmap 10.8 platform, retaining

only the segments located within the designated study area. Road segments were then reclassified according to official road grade standards or functional classifications, enabling material stock estimation by road category in subsequent analyses.

2.3. Estimation of urban road material stock

2.3.1. Bottom-up material stock framework

To quantify material stocks in an urban road system, a bottom-up MFA approach was adopted. Road material stocks were calculated for each road grade using geometric road characteristics, pavement structural layers, and material composition ratio. The calculation model is as follows:

$$S_i = f_i \rho_i [V_{total} - V_{int}] \quad (1)$$

$$V_{total} = \sum_{j=1}^n \sum_{k=1}^m L_j W_j H_{jk} \quad (2)$$

where S_i denotes the mass stock of the i -th material mixture within the urban road system; f_i represents the compaction factor assigned to the i -th mixture; ρ_i represents the dry density of i -th mixture; V_{total} represents the cumulative volume of the i -th material mixture; V_{int} denotes the overlapping volume of the i -th material mixture; n indicates the total number of road classifications ($n = 1, \dots, 5$), representing branch roads, collector roads, arterial roads, expressways, and motorways, respectively. m represents the number of structural layers in the road cross-section; L_j represents the length of roads belonging to grade j ; W_j represents the design width of roads belonging to grade j ; H_{jk} is the design thickness of the k -th structural layer for roads of grade j .

2.3.2. Correction of intersection overlap

To avoid the overestimation of material stocks resulting from the intersections of linear road segments, overlapping volumes were subtracted. Intersections were categorized into three representative types (Figure S2): T-shaped intersections, cross-shaped intersections, and multi-way intersections. The overlapping area were calculated geometrically based on road widths and intersection configuration parameters, and the corrected intersection volume were deducted from the initial road volume. Detailed explanations and schematic diagrams of the intersection overlap calculations are provided in Supplementary Information (Section 1).

For T-shaped and cross-shaped intersections, the material stock in the overlapping areas is calculated using the following formula:

$$V_{int} = \sum_{j_1=1}^n \sum_{j_2 \geq 2}^n \left\{ [\max(W_{j_1}, W_{j_2})]^2 \cdot H_{j_2} \right\} \cdot N_{j_1, j_2} \quad (3)$$

$$N_{j_1, j_2} = X_{j_1, j_2} + \frac{1}{2} T_{j_1, j_2} \quad (4)$$

where j_1 and j_2 are the functional grades of the two intersecting roads, where j_2 denotes the road with the higher classification; W_{j_1} and W_{j_2} denote the design widths of the respective intersecting roads; H_{j_2} represents the pavement thickness for road grade j_2 ; N_{j_1, j_2} represents the number of intersection nodes between roads of grade j_1 and j_2 ; T_{j_1, j_2} refers to the number of T-shaped intersections; X_{j_1, j_2} refers to the number of cross-shaped intersections.

For multi-way intersections, the material stock in the overlapping areas is calculated as follows:

$$V_{int} = \sum_{m=1}^o \pi (R_m^2 - r_m^2) \cdot H_{\max\{j_1, j_2, \dots, j_n\}} \quad (5)$$

$$r_m = \frac{\theta^2}{127(\mu \pm \theta)} - \frac{b_i}{2} \quad (6)$$

where R_m represents the area of the outer circle; r_m represents the area of the inner circle; o denotes the total number of intersecting roads; θ refers to the design speed of the roundabout, which should be calculated as 50%–70% of the maximum design speed among the intersecting roads; μ represents the friction coefficient, ranging from 0.14 to 0.18; θ denotes the cross slope of the pavement at the m -th intersection, ranging from 1.5% to 2%; b_i represents the width of the roundabout lane.

By integrating the material stock of each mixture with the cor-

responding mix ratio data, the total material stock of various road construction materials is calculated using the following equation:

$$M_z = \sum_{i=1}^{\sigma} S_i \times \gamma_{z,i} \quad (7)$$

where M_z represents the total material stock of the z -th material in the road system; σ denotes the total number of mixture types; γ_z denotes the proportion of the z -th material contained in the i -th type of mixture.

2.4. Driving force decomposition using the LMDI method

2.4.1. Conceptual decomposition framework

To investigate the mechanisms underlying changes in road material stocks, a multiplicative decomposition framework was constructed based on the extended Kaya identity (Kaya et al., 1989; Deng et al., 2022; Nguyen et al., 2023). By disaggregating the total material stock of an urban road system into infrastructure scale, economic activity, demographic structure, and design-related intensity variables, the decomposition model can be expressed as follows:

$$S = \frac{S}{A} * \frac{A}{L} * \frac{L}{GDP} * \frac{GDP}{P_1} * \frac{P_1}{P} * P = MI * AL * LG * GP * U * P \quad (8)$$

where S denotes road material stocks (Mt); A represents road area (m²); L is road length (m); GDP is the annual gross domestic product (CNY million); P_1 is urban population (million persons); P is total population (million persons); MI represents material stock per unit area and reflects the material utilization intensity effect; AL reflects the road width effect; LG denotes road length intensity and represents the road length effect; GP denotes GDP per urban population and reflects the economic effect; U denotes urbanization rate (P_1/P) and reflects the urbanization effect.

2.4.2. Additive LMDI decomposition

Based on the above expression, the change in material stocks between year $t-1$ and t is decomposed using the additive Logarithmic Mean Divisia Index (LMDI) method:

$$\Delta S = S_t - S_{t-1} = \Delta S_{MI} + \Delta S_{AL} + \Delta S_{LG} + \Delta S_{GP} + \Delta S_U + \Delta S_P \quad (9)$$

$$\Delta S_{MI} = \frac{S_t - S_{t-1}}{\ln S_t - \ln S_{t-1}} \ln \frac{MI_t}{MI_{t-1}} \quad (10)$$

$$\Delta S_{AL} = \frac{S_t - S_{t-1}}{\ln S_t - \ln S_{t-1}} \ln \frac{AL_t}{AL_{t-1}} \quad (11)$$

$$\Delta S_{LG} = \frac{S_t - S_{t-1}}{\ln S_t - \ln S_{t-1}} \ln \frac{LG_t}{LG_{t-1}} \quad (12)$$

$$\Delta S_{GP} = \frac{S_t - S_{t-1}}{\ln S_t - \ln S_{t-1}} \ln \frac{GP_t}{GP_{t-1}} \quad (13)$$

$$\Delta S_U = \frac{S_t - S_{t-1}}{\ln S_t - \ln S_{t-1}} \ln \frac{U_t}{U_{t-1}} \quad (14)$$

$$\Delta S_P = \frac{S_t - S_{t-1}}{\ln S_t - \ln S_{t-1}} \ln \frac{P_t}{P_{t-1}} \quad (15)$$

where ΔS denotes the change in material stocks; S_t and S_{t-1} represent the material stocks in years t and $t-1$, respectively. ΔS_{MI} , ΔS_{AL} , ΔS_{LG} , ΔS_{GP} , ΔS_U , and ΔS_P denote the effects of material utilization intensity, road width, road length, economic, urbanization, and population, respectively, on the change in material stocks.

2.4.3. Conceptual grouping of driving factors

For analytical clarity, the six effects are further categorized into three conceptually distinct groups. This classification allows the framework to examine whether material stock growth in urban road systems is primarily driven by infrastructure scale expansion, moderated by internal efficiency improvements, or influenced by socio-demographic structural transitions.

(1) Scale-driven effects

Scale-driven effects increase total infrastructure volume and are typically associated with network expansion (road length effect (ΔS_{LG})) and investment growth (economic effect (ΔS_{GP})). These factors directly expand the physical scale of the road system or reflect macroeconomic expansion.

(2) Design-related efficiency effects

Design-related efficiency effects reflect design standards, cross-section configuration, and material allocation practices, including road width effect (ΔS_{AL}) and material utilization intensity effect (ΔS_{MI}). These factors modify material input per unit infrastructure without changing system scale.

(3) Socio-structural effects

Socio-structural effects capture demographic and structural transitions influencing infrastructure demand, including urbanization effect (ΔS_U) and population effect (ΔS_P). These effects represent broader socio-economic structural dynamics rather than direct physical or design changes.

2.5. Life cycle assessment

2.5.1. Goal and scope definition

The life cycle assessment (LCA) component of the framework aims to quantify environmental impacts associated with the total in-use urban road material stocks in each study year across defined life cycle stages. The assessment adopts a stock-based perspective; therefore, the results represent the environmental burdens embodied in the accumulated road infrastructure stock at each time point rather than the impacts associated only with annual road construction activities. Accordingly, the functional unit was defined as the entire urban road system within a specified administrative boundary for a given year. The system boundary followed a cradle-to-construction framework comprising standard modules A1–A5, which accounts for upstream material production (A1–A3: raw material supply, transport to the manufacturing facility, and the manufacturing process), transportation to the construction site (A4), and on-site road construction activities (A5).

2.5.2. Life cycle inventory

The material quantities used in the life cycle inventory were derived from the total in-use road material stock estimated by the MFA module for each study year, rather than from annual material inflows associated with newly constructed roads. Material inputs for Modules A1–A3 were derived from the MFA outputs (Section 2.3), where the material composition ratios in Table S4 were used to disaggregate each pavement mixture from the outputs into constituent materials, including mineral powder. The corresponding production energy use was quantified by mapping material production processes to the Highway Engineering Budget Quota and Highway Engineering Machinery Shift Quota. For module A4, transportation inventories were calculated by combining MFA-derived material quantities with region-specific freight distances. For module A5, machinery energy consumption was calculated by decomposing the construction activities of each pavement structural layer in accordance with the aforementioned national budget and machinery shift quotas. These foreground inventory flows were then linked to appropriate background datasets in the Ecoinvent 3 database for materials, energy inputs, and transport processes, with detailed dataset names provided in the Supplementary Information (Tables S6–S8). To improve geographic representativeness, each material and energy flow was matched with the most appropriate regional inventory dataset available in Ecoinvent 3, for example, electricity consumption was linked to the Chinese electricity market dataset.

2.5.3. Life cycle impact assessment

Environmental impacts were evaluated using the ReCiPe 2016(H) impact assessment method, which provides both midpoint and endpoint indicators (Huijbregts et al., 2017). The midpoint impacts include impact categories such as global warming potential, ozone depletion potential, and ionizing radiation potential, while the endpoint impacts include human health, ecosystem quality, and resource scarcity. The assessment was implemented using SimaPro 9.0.0.48 software platform to ensure comparability, consistent with the methodological setup adopted in previous road infrastructure LCA studies, such as Agheh et al. (2025).

2.5.4. Interpretation

The interpretation focuses on identification of dominant life cycle stages, comparison of material-specific environmental contributions, and temporal trend analysis of aggregated impacts. The results are used to examine whether changes in material stocks translate into proportional changes in environmental impacts, thereby linking physical stock accumulation with environmental consequences.

3. Case application: shanghai

3.1. Study area characteristics

Shanghai is located on the eastern coast of China and covers a total administrative area of 6340.5 km². The municipality consists of 16 administrative districts, including seven central urban districts and nine outer suburban districts (Shanghai Statistical Yearbook, 2024). As one of China's most economically developed megacities, Shanghai has experienced sustained infrastructure investment and spatial expansion over the past decade. The urban road system comprises multiple hierarchical road grades, including motorway, expressway, arterial road, collector road, and branch road. The city provides a representative case for applying the proposed framework due to its high urbanization level, complex road hierarchy, and continuous infrastructure upgrading.

3.2. Data sources and parameterization

To implement the proposed framework, spatial, structural, socio-economic, and life cycle inventory data were compiled for Shanghai for the period 2014–2023.

3.2.1. Urban road network data

OSM historical data snapshots corresponding to January 1 of each year from 2014 to 2023 were extracted and used as annual reference datasets to represent road network conditions for each year. The OSM-derived datasets, obtained from OpenStreetMap (OSM) through the Geofabrik download service (<http://download.geofabrik.de/>) in shapefile format compatible with ArcMap 10.8, had been used in previous infrastructure stock studies (Wiedenhof et al., 2024b). Although OSM completeness may vary across regions and time periods, previous research indicates that OSM-based datasets in China tend to show higher completeness in built-up urban areas (Wang et al., 2020). Administrative boundary data were used to clip the road network within Shanghai's jurisdiction. All spatial data were projected into the same coordinate system to ensure geometric consistency.

According to the Code for Design of Urban Road Engineering (CJJ37–2012, 2016 edition) (Ministry of Housing and Urban-Rural Development of the People's Republic of China, 2016), urban roads in China are classified into four grades: expressway, arterial road, collector road, and branch road. Each road grade differs in terms of status and traffic function. Considering the presence of motorways within the study area, these are also included as an additional category in this study. To ensure consistency between the OSM road classification and the Chinese road hierarchy, OSM tags were reclassified by function, with motorway as motorway, trunk as expressways, primary and secondary as arterial

roads, tertiary as collector roads, and residential and unclassified as branch road.

Road width values were assigned based on the Code for Design of Urban Road Engineering (CJJ37–2012, 2016 edition) (Ministry of Housing and Urban-Rural Development of the People's Republic of China, 2016) and official road design documents. Additionally, the number of motor vehicle lanes for each road grade was identified using open-source data from Google Maps and Baidu Maps, as shown in Table S1. For roads lacking explicit width information, the average width of the corresponding road grade was applied.

Road network intersections were identified from the processed road network dataset using topological analysis of network nodes and subsequently classified into T-shaped, cross-shaped, and multi-way intersections. The corresponding intersections areas were quantified based on geometric relationships derived from road width parameters.

3.2.2. Urban road structure and material composition

Typical pavement structures composition of roads by grade as illustrated in Fig. 2 were defined according to the 15MR201 Urban Road Asphalt Pavement Standard Atlas and related technical specifications (China Building Standard Design & Research Institute, 2015). The composition ratio of each mixture type in road structure is presented in Table S4.

3.2.3. Socio-economic data

Population, GDP (at constant 2020 prices), and urbanization rate data were obtained from the Shanghai Statistical Yearbook (2015–2024) and China Statistical Yearbook (2015–2024). The urban population was determined by multiplying the total population by the urbanization rate, consistent with the definition of urbanization rate as the proportion of urban population to total population (Wang and Zhang, 2022).

3.2.4. Transportation and energy inventory data

Energy inventory data for material production and construction stages were derived from the Highway Engineering Budget Quota and Highway Engineering Machinery Work Quota. Freight transportation was modeled using representative 16–32 ton trucks from the Ecoinvent 3 database, with an assumed average transport distance of 36 km (Rousseau et al., 2022).

3.2.5. Data validation and reliability assessment

To ensure the robustness of the spatial and geometric data used in the material stock estimation, validation procedures were conducted for both road length and road width parameters. For road length validation, the extracted motorway length from the processed OSM road network was compared with official statistics reported in the Shanghai Statistical Yearbook (2015–2024) for each year from 2014 to 2023. The year-by-year comparison is provided in Supplementary Information (Section 6, Table S9). The mean relative error across the study period was 16.48%. This deviation falls within a reasonable range when compared with similar studies. For example, (Wiedenhöfer et al., 2024a) reported deviations of +135% for Great Britain and +62% for Ireland when comparing OpenStreetMap data with national statistics, attributing discrepancies to differences in road classification standards and contributor practices. For road width, taking Huyi Highway in Jiading District, Shanghai as an example, it is classified as an arterial road with six lanes. The measured width from Google Maps is 21.65 m, whereas this study assumes a width of 22.5 m, yielding a deviation of 3.92%. As another example, Chenggu Road is classified as a collector road with two lanes. The Google Maps measured width is 7.24 m, while this study assumes a width of 7.75 m, resulting in a deviation of 7.04%. This level of discrepancy is considered acceptable, as Karjalainen et al. (2024) reported relative errors of approximately 7–10% in roadway width estimation and concluded that such accuracy is adequate. Considering both length and width validation results, the geometric inputs used for material stock estimation are deemed sufficiently robust for annual stock quantification and decomposition analysis.

4. Results

4.1. Temporal evolution of road material stocks

The total and material specific road material stocks in Shanghai from 2014 to 2023 is illustrated in Fig. 3a. Over the study period, the total material stock increased continuously from approximately 416.75 Mt in 2014 to 810.61 Mt in 2023, corresponding to an average annual growth rate of 7.67%. The overall trajectory can be divided into three distinct phases comprising a period of rapid accumulation (2014–2016), a phase of steady consolidation (2017–2020), and a renewed acceleration after 2021. This recent uptick underscores an intensified investment in urban

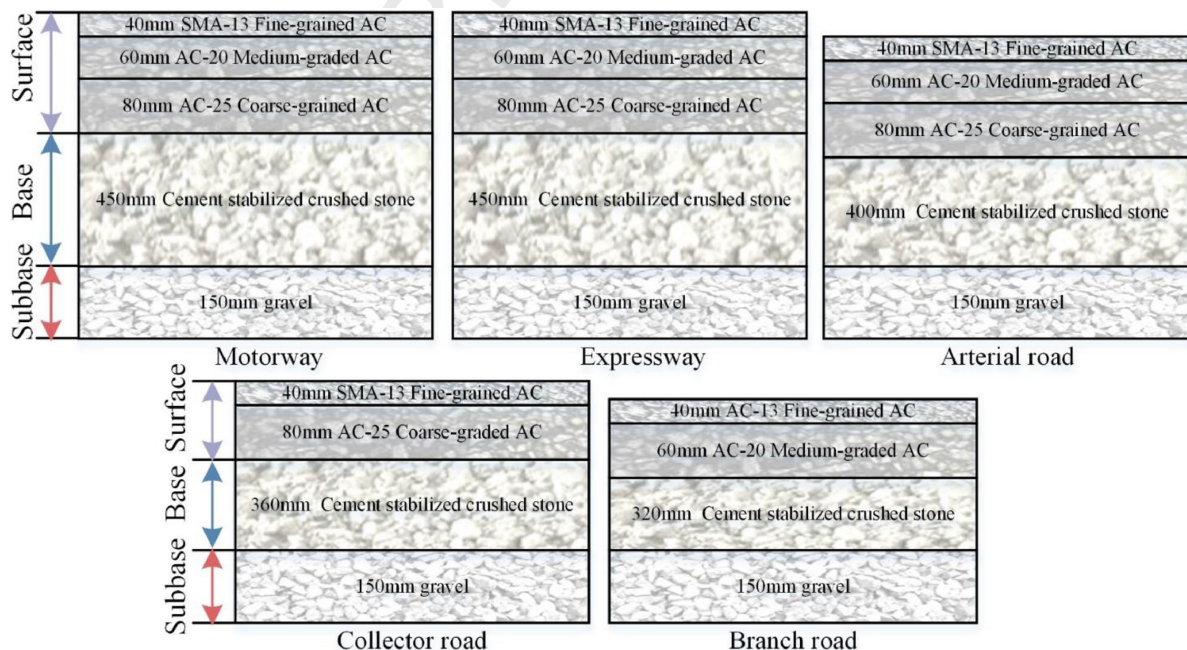


Fig. 2. Typical pavement structures composition of roads by grade.

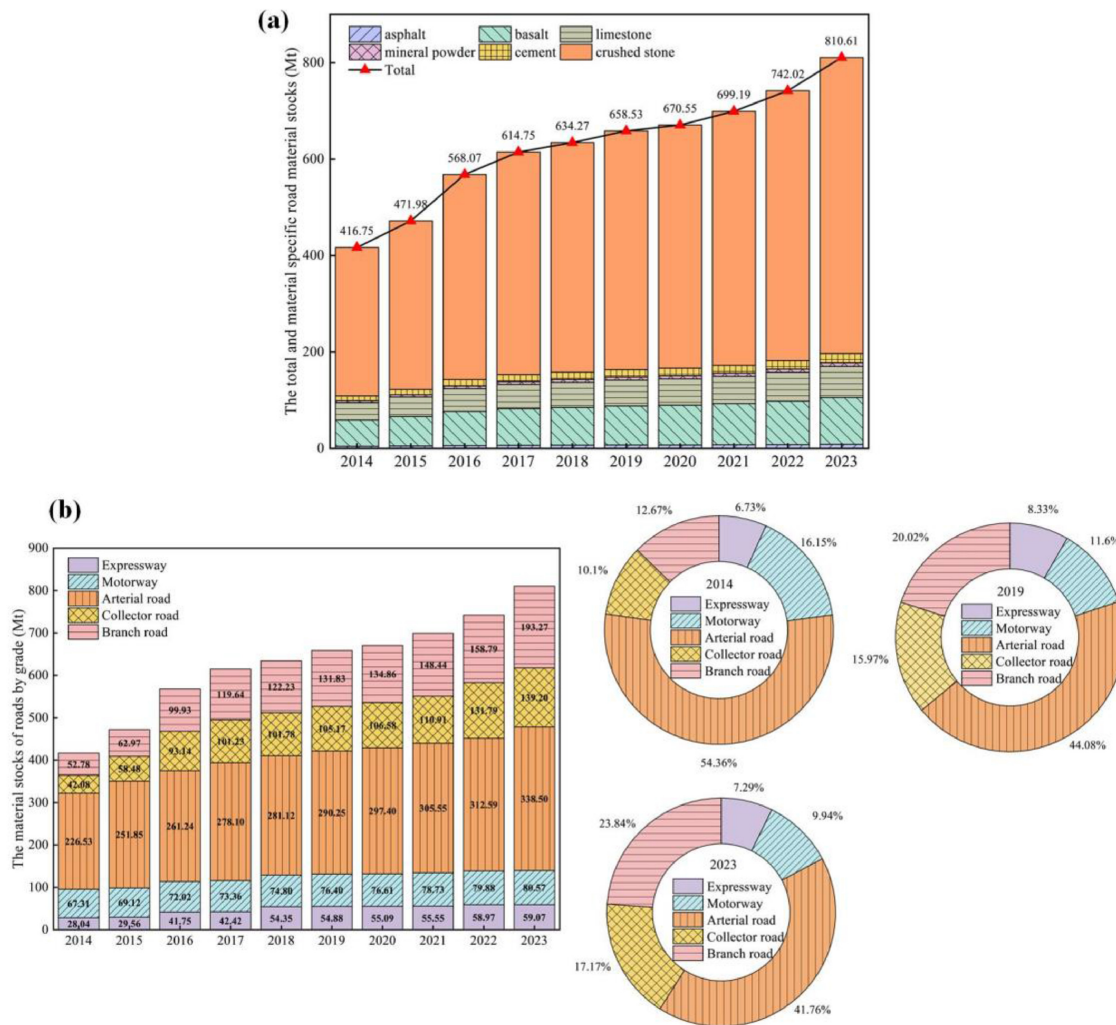


Fig. 3. Composition and distribution of road material stocks by material type and road grade. Subfigure (a) presents the total and material specific road material stocks. Subfigure (b) illustrates road material stocks by road grade in absolute values and percentage shares.

transportation infrastructure to support Shanghai's evolving metropolitan functions. In terms of material composition, crushed stone and basalt dominated the total material stock, followed by cement-based materials and asphalt. As shown in Fig. 3a, the stock of crushed stone exhibits a persistent upward trend and accounts for the majority of the absolute increase in total material stocks, with a growth rate of 99.33% from 2014 to 2023. In contrast, asphalt materials represent a smaller proportion of the total stock and show more moderate growth of 88.17% over time. Despite differences in growth rates among material types, the overall composition of road material stocks remains relatively stable during the study period, with mineral aggregates consistently forming the core of Shanghai's road infrastructure.

The road material stocks by road grade in absolute values and percentage shares is shown in Fig. 3b. For visual clarity, three representative years, 2014, 2019, and 2023, are shown to illustrate the overall evolution during the study period. Arterial roads consistently accounted for the largest share, maintaining a proportion above 40% despite a gradual decline, which underscores their central role in the urban traffic network. Notably, the proportion of branch roads showed a continuous upward trend, rising from 12.67% in 2014 to 23.84% in 2023. In contrast, the proportion of motorway declined from 16.15% to 9.94%, while that of expressway first increased and then decreased. These results indicate that urban road construction resources have gradually shifted toward medium-grade and lower-grade roads.

4.2. Spatial distribution of road material stocks

Shanghai consists of 16 administrative districts, including seven central urban districts and nine outer suburban districts. Fig. 4a compares the absolute road material stocks across all districts from 2014 to 2023. In absolute terms, road material stocks were generally higher in outer districts than in central urban districts throughout the study period. In 2014, the outer districts collectively contained 341.78 Mt of road material stocks, compared with 74.97 Mt in the central districts. By 2023, these values had increased to 696.87 Mt and 113.74 Mt, respectively. Among all administrative districts, Pudong New Area consistently exhibited the highest absolute road material stock, which is partly attributable to its large geographical area and extensive road infrastructure.

However, direct comparisons of absolute material stocks across districts may be influenced by differences in geographical area. Road material stock density was therefore calculated as the total road material stock per unit of administrative area (Mt/km^2), as shown in Fig. 4b. For the group-level comparison, stock density was calculated as the total material stock of the central or outer districts divided by the corresponding total administrative area. In contrast to the pattern observed for absolute stocks, the area-normalized results show substantially higher road material stock densities in central urban districts. The road material stock density of the central districts increased from approximately $0.26 \text{ Mt}/\text{km}^2$ in 2014 to $0.39 \text{ Mt}/\text{km}^2$ in 2023, whereas that of the outer districts increased from 0.06 to $0.12 \text{ Mt}/\text{km}^2$. By 2023, the material

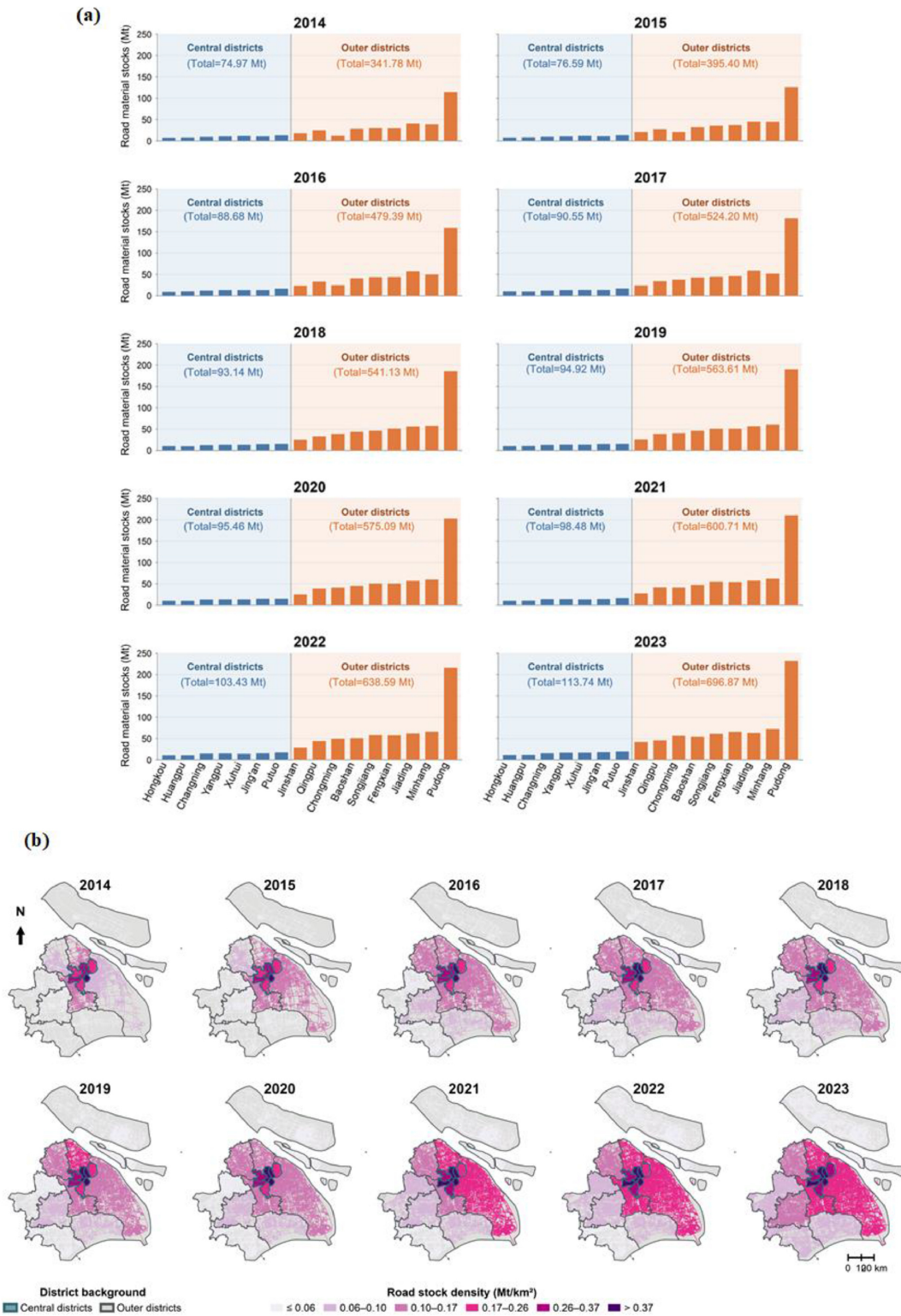


Fig. 4. Spatial distribution of road material stocks and road material stock density in Shanghai by district. Subfigure (a) presents the total road material stocks in Shanghai by district (unit: Mt). Subfigure (b) depicts the spatial distribution of road material stock density in Shanghai (unit: Mt/km^2).

stock density of central districts was more than three times that of outer districts.

These results reveal a clear distinction between absolute stock magnitude and area-normalized stock density. Outer districts contain larger absolute quantities of road materials, partly reflecting their greater geographical extent and extensive road network, whereas central districts contain substantially more road material stock per unit land area.

4.3. Driving factors of road material stock growth

The decomposition contributions of different driving factors are illustrated in Fig. 5a. For analytical clarity, the driving factors are grouped into three categories: scale-driven effects (road length and economic effects), design-related efficiency effects (road width and material utilization intensity), and socio-structural effects (population and urban-

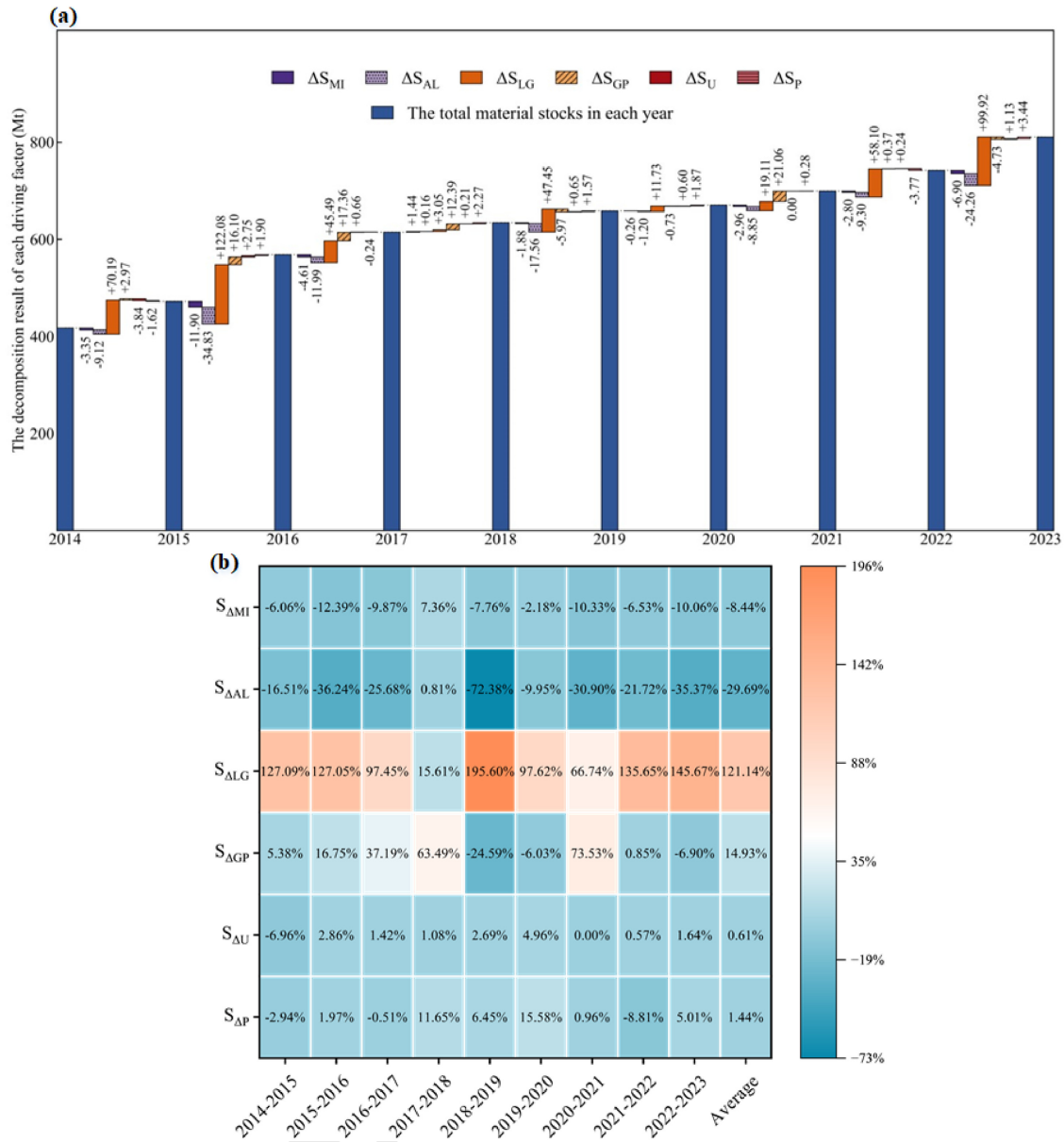


Fig. 5. Decomposition of the driving factors of road material stock growth. Subfigure (a) presents the decomposition contributions of different driving factors. Subfigure (b) illustrates the proportional decomposition contributions of different driving factors.

Notes: ΔS_{MI} , ΔS_{AL} , ΔS_{LG} , ΔS_{GP} , ΔS_U and ΔS_P represent the material utilization intensity effect, road width effect, road length effect, economic effect, urbanization effect, and population effect, respectively. S_{AMI} , S_{DAL} , S_{ALG} , S_{AGP} , S_{AU} , and S_{AP} represent the proportional contributions of the material utilization intensity effect (ΔS_{MI}), road width effect (ΔS_{AL}), road length effect (ΔS_{LG}), economic effect (ΔS_{GP}), urbanization effect (ΔS_U), and population effect (ΔS_P), respectively.

ization). Among the scale-driven effects, the road length effect maintained a positive contribution throughout the entire study period. Its annual contribution ranged from 3.05 Mt to 122.08 Mt, indicating a consistently positive association between network expansion and material stock growth. The economic effect acted as a secondary positive driver in most years. However, negative contributions were observed during 2018–2020 and 2022–2023, indicating that economic effect did not consistently contribute positively to material stock growth during these periods.

The socio-structural effects exhibited comparatively smaller magnitudes. The population effect turned negative during 2014–2015, 2016–2017, and 2021–2022, suggesting that occasional population declines temporarily restrained infrastructure demand, thereby inhibiting the growth of material stock. The urbanization effect shifted to a consistently positive contribution starting in 2015, although its overall impact on promoting stock growth remained comparatively limited, with the

majority of its annual values remaining below 1 Mt.

By contrast, design-related efficiency effects generally exerted offsetting contributions to material stock growth. The road width effect was identified as the dominant offsetting design-related factor, indicating that changes in the network-wide average road width (A/L) partially offset material stock growth. This pattern is consistent with the evolving road grade structure observed in Section 4.1, where the contribution of relatively narrower branch roads increased while that of wider motorways declined over the study period. Similarly, the material utilization intensity effect was negative in all years except for 2017–2018, suggesting that variations in material input per unit road area moderated overall stock growth.

To provide a more detailed elucidation of the driving mechanisms underlying each effect, the proportional decomposition contributions of different driving factors are presented in Fig. 5b. Over the entire study period, the road length effect exhibited a dominant average contribu-

tion of 121.14%, exceeding the total observed net growth. The economic effect showed an average contribution of 14.93%, although its annual values fluctuated. Design-related efficiency factors exerted offsetting effects, with average contributions of -29.69% for the road width effect and -8.44% for the material utilization intensity effect. In comparison, the population effect and urbanization effect contributed only 1.44% and 0.61% on average, respectively.

4.4. Life cycle environmental impacts associated with road material stocks

The cradle-to-construction environmental impacts were assessed based on the total in-use road material stock in each study year rather than annual material inflows. Accordingly, the temporal results represent changes in the stock-based environmental burdens associated with the accumulated road infrastructure. Fig. 6a presents the proportional contributions of cradle-to-construction stages and single score environmental impacts (normalized and weighted ReCiPe 2016(H) endpoint environmental impacts) associated with road material stocks. Environmental impacts are expressed as points, a dimensionless indicator representing aggregated environmental burdens after normalization and weighting across multiple impact categories. The single score increased continuously from 7.66×10^8 points in 2014 to 1.50×10^9 points in 2023. A rapid growth was observed between 2014 and 2016, including a sharp increase of 20.69% between 2015 and 2016. After 2016, the single score continued to rise, although the annual growth rate varied across years. Throughout the study period, the distribution of environmental impacts associated with road material stocks among cradle-to-construction stages remained highly stable, with material production stage consistently accounting for approximately 88% of the total im-

pacts, followed by transportation stage at around 10–11%. In contrast, the construction stage contributed only about 1.3%.

The share of cradle-to-construction environmental impacts associated with road material stocks from different road construction materials is presented in Fig. 6b. Crushed stone and cement accounted for the largest shares of total environmental burdens. In contrast, asphalt and basalt contributed smaller proportions, accounting for an average of 9.82% and 7.60%, respectively. Consequently, decarbonization and resource-saving efforts should target bulk material substitution.

Proportion of midpoint environmental impacts associated with road material stocks across cradle-to-construction stages is illustrated in Fig. 6c. The material production stage contributed the largest share across all midpoint categories, generally exceeding 80% of the total impacts. This dominance was particularly evident in categories such as water consumption, land use, freshwater eutrophication, and mineral resource scarcity. These findings suggest that the extraction, processing, and manufacturing of materials processes are highly resource-intensive while generating considerable environmental impacts. The transportation stage exhibited higher relative contributions in terrestrial ecotoxicity and stratospheric ozone depletion, accounting for 42.50% and 23.18%, respectively. This highlights the considerable effects of fuel consumption and related emissions during long-distance transport on ecosystems and the atmospheric environment. Additionally, its contributions to fossil resource scarcity and human non-carcinogenic toxicity were comparatively higher than in other midpoint categories. The construction stage contributed the smallest share across all midpoint impact categories, with most values below 2%, indicating its relatively negligible role compared to the production and transportation phases.

Proportion of endpoint environmental impacts associated with road material stocks across cradle-to-construction stages is presented in

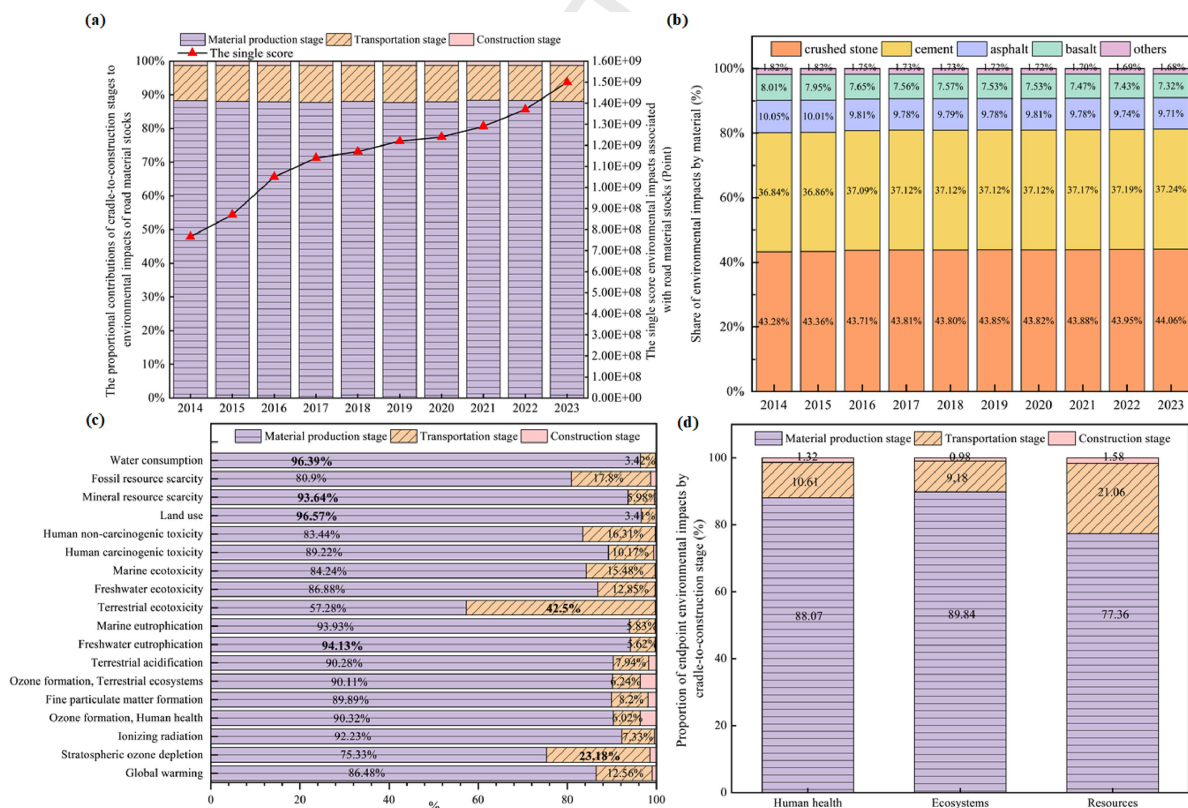


Fig. 6. Environmental impacts associated with road material stocks across cradle-to-construction stages. Subfigure (a) presents the proportional contributions of cradle-to-construction stages and single score environmental impacts associated with road material stocks. Subfigure (b) illustrates the share of cradle-to-construction environmental impacts associated with road material stocks from different road construction materials. Subfigure (c) depicts the proportion of midpoint environmental impacts associated with road material stocks across cradle-to-construction stages. Subfigure (d) shows the proportion of endpoint environmental impacts associated with road material stocks across cradle-to-construction stages.

Fig. 6d. The material production stage dominated all three endpoint categories, namely human health, ecosystems, and resources, accounting for 88.07%, 89.84%, and 77.36%, respectively. The transportation stage contributed 21.06% to resource-related impacts, a higher proportion than its contributions to human health and ecosystem damage. This finding indicates the substantial potential of the transportation process for improving resource efficiency. The construction stage accounted for <2% in all endpoint categories, further suggesting that its overall environmental burden was relatively limited.

5. Discussion

This study set out to address a central but unresolved question in urban road material metabolism: whether the continued growth of road material stocks in mature megacities is primarily an inevitable outcome of scale-driven demand or whether internal infrastructure design and efficiency choices can meaningfully moderate material accumulation and its environmental consequences. By integrating MFA, LMDI decomposition, and LCA for Shanghai from 2014 to 2023, the results provide empirical evidence that both mechanisms coexist, but operate in fundamentally different ways. While network extension and economic expansion remain the dominant forces driving absolute material stock growth, persistent negative effects associated with road width and material intensity indicate the presence of design-level efficiency gains that partially offset, but do not reverse, the overall growth trajectory.

In the following discussion, we interpret the observed spatiotemporal patterns of material stocks, driving factors, and environmental impacts through the lens of this scale-efficiency tension. We first examine the extent to which scale-driven factors dominate material accumulation in Shanghai, and then assess the significance of internal design-related efficiency signals. We subsequently discuss how these dynamics translate into life-cycle environmental impacts and what they imply for resource governance in mature urban road systems.

5.1. Scale-driven growth versus efficiency signals

The LMDI decomposition shows that the road length effect consistently exerts the largest positive contribution to material stock growth through the study period, often exceeding the magnitude of total stock change. This pattern underscores the material-intensive nature of extending physical transport infrastructure.

The economic effect represents a secondary but persistent source of growth. Although its contribution fluctuated over time, it remained predominantly positive. This pattern suggests that efficiency gains embedded in the road system are often insufficient to fully counterbalance the upward pressure associated with economic growth, a finding that is consistent with previous studies on infrastructure material stocks in both developing and developed contexts (Zhang et al., 2019; Tareq et al., 2025). In contrast, population and urbanization effects play a relatively minor role in Shanghai, with the population effect showing offsetting contributions during several periods. This divergence from patterns observed in rapidly growing or less mature urban systems suggests that, in a highly urbanized megacity such as Shanghai, road material stock dynamics may be less strongly associated with demographic growth and more strongly influenced by economic activity and road network expansion. Despite the limited contribution of demographic factors, road material stocks continued to increase substantially from 2014 to 2023, indicating that the road system did not exhibit signs of stock stabilization during the study period. This continued accumulation suggests that, during the study period, road network expansion and economic activity played a more important role in stock accumulation than demographic change.

5.2. Design-level efficiency gains and their significance

The LMDI decomposition results also reveal that internal design characteristics of Shanghai's road system exert a measurable influence on material stock dynamics. Specifically, the road width effect was identified as the dominant inhibiting factor over the study period, indicating that changes in the network-wide average road width (A/L) partially offset material stock growth. Similarly, the material utilization intensity effect remains predominantly negative, suggesting changes in materials input per unit road area generally moderated stock growth. It should be noted that road width and material utilization intensity are not short-term adjustable variables at the project level, but are determined by road hierarchy, design standards, cross-sectional configuration, pavement structure, and material specifications. The negative road width effect does not imply the physical narrowing or demolition of existing roads. Rather, it reflects a network-level change in average road width associated with the evolving road grade structure. Accordingly, the LMDI results capture design-related efficiency effects embedded in the evolution of the road system, rather than suggesting that existing roads can be directly narrowed or redesigned in practice. These effects mainly reflect medium- to long-term structural and design governance effects, relevant to new construction, upgrading, reconstruction, and design standard revision.

Comparable efficiency signals have been observed in other contexts. For instance, Deng et al. (2022) and Long et al. (2024) utilized the LMDI model to quantify the efficiency of urban and building stocks, where efficiency is reflected in the relationship between material stock accumulation and economic growth. However, the present study extends these insights by explicitly quantifying the role of internal road design characteristics at the urban scale, rather than inferring efficiency improvements indirectly.

5.3. Spatial contrasts in road material stock magnitude and density

The spatial analysis reveals an important distinction between the absolute magnitude and area-normalized density of road material stocks in Shanghai. Although outer districts generally contain larger absolute material stocks, central districts exhibit substantially higher material stock densities. This finding demonstrates that absolute material stock alone does not fully characterize the spatial intensity of infrastructure accumulation, particularly when administrative units differ considerably in geographical area.

This distinction also provides a more appropriate basis for comparison with previous spatial material stock studies. However, direct comparison should be made cautiously because previous studies adopted different spatial units and indicators. Mollaei et al. (2021) represented road material stocks at the individual road-segment level and observed a denser road network in downtown Kitchener and Waterloo, although these central roads were predominantly local roads with relatively lower material intensities. Miatto et al. (2021) mapped road material stocks in Hanoi using a 500 m grid and further examined spatial stock density, demonstrating that the observed spatial pattern depends on both road-network configuration and the spatial unit of analysis. Zhao et al. (2023) similarly reported spatial heterogeneity in urban road material stocks in Kunming. These studies indicate that road-network density, absolute material stock, and area-normalized stock density describe different dimensions of spatial material accumulation. Therefore, apparent differences among cities should not be interpreted solely as contrasting central-outer patterns, because spatial resolution, accounting units, road classifications, and system boundaries can substantially influence the observed distributions.

The magnitude of road material stocks in Shanghai also differs from previous studies. Han et al. (2018) reported 23.56 Mt for Shanghai in 2010, substantially lower than our estimate, mainly due to their focus on high-grade highways only. Similarly, Zhao et al. (2023) reported 552 Mt for Kunming in 2020, which is lower than the 670.55 Mt estimated in

this study, while Yu et al. (2021) reported 112 Mt for the urban core area of Nanjing in 2018, compared with 634.27 Mt in Shanghai. These differences largely reflect variations in system boundaries, road classification, and pavement material assumptions. These comparisons highlight the strong sensitivity of road material stock estimates to system boundaries, spatial coverage, road classifications, and pavement structure assumptions, and therefore caution against direct comparison of absolute stock values across cities.

The results therefore suggest the coexistence of two spatial processes: intensive material occupation per unit land area in the urban core and continued absolute stock accumulation in peripheral districts. This pattern reflects the different stages and spatial forms of road infrastructure development within Shanghai. Central districts already contain dense road infrastructure within relatively limited land area, resulting in high material stock density, whereas outer districts accommodate larger road networks over much greater geographical areas and continue to account for most of the absolute stock increase. Rather than indicating a simple outward concentration of road materials, the results reveal the coexistence of high infrastructure intensity in the urban core and continued material accumulation in peripheral areas. This distinction provides a more nuanced understanding of the spatial metabolism of Shanghai's road infrastructure and highlights the importance of considering both absolute stock magnitude and area-normalized density in urban material stock analysis.

5.4. Environmental implications of material accumulation

Despite observable efficiency gains at the design level, the life cycle assessment results indicate that absolute environmental impacts associated with road material stocks by cradle-to-construction stages continue to increase over time. The single score environmental impact rises steadily from 2014 to 2023, mirroring the overall growth in material stocks. This trend demonstrates that, while design-related efficiency effects associated with road width and material intensity may moderate the escalation of environmental impacts, they cannot fully offset the effects of continued network expansion and economic growth.

The dominance of the material production stage across most midpoint and endpoint impact categories further highlights the structural nature of environmental pressures associated with road infrastructure. Crushed stone and cement, which constitute the bulk of road material stocks, emerge as the primary contributors to environmental burdens, particularly in categories related to resource depletion, water consumption, and ecosystem impacts. This finding aligns with previous LCA studies of road and building infrastructure, which consistently identify material production as the main source of life-cycle environmental impacts (Chaudhary et al., 2024; Chen et al., 2025; Hu et al., 2023).

Importantly, the environmental implications of efficiency gains are therefore asymmetric. While design-related efficiency effects associated with road width and material intensity may moderate the escalation of environmental impacts, they cannot fully offset the effects of continued network expansion and economic growth. This asymmetry underscores the risk of relying solely on efficiency-oriented strategies to address the environmental consequences of urban road material accumulation.

5.5. Governance implications and limits of efficiency-oriented strategies

The coexistence of strong scale-driven growth and measurable design-level efficiency gains has important implications for resource governance in mature urban road systems. On the one hand, the persistent negative effects associated with road width and material intensity suggest that efficiency-oriented design standards can play a meaningful role in moderating material accumulation. These internal factors represent medium- to long-term leverage points for planners and engineers, as they are shaped by road hierarchy, design standards, technical specifications, and decisions related to new construction, upgrading, and reconstruction. On the other hand, the dominance of economic and net-

work expansion effects highlights the structural limits of efficiency-based interventions. As long as road infrastructure continues to be expanded as a response to economic development objectives, efficiency gains are likely to function as relative improvements rather than transformative solutions. In this sense, efficiency strategies may reduce the environmental intensity of road systems but cannot, on their own, reverse absolute growth in material stocks or environmental impacts.

These findings point to the need for a more integrated governance approach that combines efficiency-oriented design with broader demand-side and system-level strategies, such as prioritizing multimodal transport, optimizing network utilization, and reassessing the necessity of continued physical expansion. From a resource perspective, failing to address scale-driven demand risks reinforcing long-term material lock-in, whereby accumulated road materials commit cities to sustained resource consumption and environmental burdens for decades.

6. Conclusions

This study developed an integrated MFA-LMDI-LCA analytical framework to systematically investigate the spatiotemporal evolution, driving mechanisms, and environmental consequences of the total in-use urban road material stocks, using Shanghai as a representative mature megacity case. By explicitly distinguishing between scale-driven, design-related efficiency, and socio-structural effects, this study enables a clearer understanding of the relative roles of infrastructure expansion and internal optimization in shaping material accumulation. The main findings are summarized as follows.

Road material stocks in Shanghai exhibit a pronounced spatiotemporal evolution pattern, increasing from 416.75 Mt in 2014 to 810.61 Mt in 2023, with rapid growth in the early years, moderated expansion during 2017–2020, and renewed growth after 2021. Spatially, material stocks remained lower in central districts and higher in outer districts, whereas road material stock density was higher in central districts.

This sustained growth pattern of material stocks is shaped by its underlying driving mechanisms, which are predominantly scale-related. The road length effect was the dominant contributor, while the economic effect provides additional positive contributions in most years. In contrast, design-related efficiency factors, including road width effect and material utilization intensity, generally exerted negative effects, partially offsetting stock growth. Socio-structural effects, such as population and urbanization, played comparatively minor roles.

The continued expansion of material stocks further translates into increasing environmental burdens. Environmental impacts associated with the total in-use road material stocks, expressed as the normalized and weighted ReCiPe 2016 (H) endpoint single score across cradle-to-construction stages, increased from 7.66×10^8 points in 2014 to 1.50×10^9 points in 2023. Among construction material, crushed stone and cement contributed the largest shares. Across cradle-to-construction stages, material production dominated most midpoint and all endpoint impact categories, followed by transportation, while construction stage contributed the least.

Despite these findings, certain limitations remain to be addressed in future research. The current spatial resolution is constrained by data availability, as several key inputs are only available at the district level, limiting the ability to capture finer-scale heterogeneity. Additionally, although validation was conducted using available statistical data and selected case comparisons, the lack of high-resolution and independent datasets restricts more rigorous cross-validation. Furthermore, the estimation of road material stocks assumes that pavement structures and material compositions remain constant within each road grade over time, which may overlook temporal and local variations in engineering practice. In addition, maintenance activities were excluded from the LCA scope, which limits the comprehensiveness of the environmental impact assessment across the full life cycle. Future research could address these limitations by integrating higher-resolution spatial data, incorporating independent datasets for validation, introducing temporal

dynamics in pavement structures and material compositions, accounting for maintenance-related material flows and emissions, and linking infrastructure material accumulation more directly with transport demand indicators.

CRedit authorship contribution statement

Dan Chong: Writing – review & editing, Validation, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Jinzha Zhang:** Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Wei Sheng:** Methodology, Conceptualization. **Siqi Jia:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Shiyuan Li:** Software, Investigation, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.resconrec.2026.109180](https://doi.org/10.1016/j.resconrec.2026.109180).

Data availability

Data will be made available on request.

References

- Aghel, S., Gholamalifard, M., Bahramifar, N., Ahmadi, B., Roshanravan, B., 2025. A comparative life cycle assessment (LCA) of the construction of the third section of the Tehran-shomal freeway: largest road construction project in Iran. *Clean. Responsible Consum.* 18, 100294. <https://doi.org/10.1016/j.clrc.2025.100294>.
- Cai, B., Baumgart, A., Haberl, H., Wiedenhofer, D., Fang, S., Shao, Z., 2024. Mapping material stocks in buildings and infrastructures across the Beijing-Tianjin-Hebei urban agglomeration at high-resolution using multi-source geographical data. *Resour. Conserv. Recycl.* 205, 107561. <https://doi.org/10.1016/j.resconrec.2024.107561>.
- Chaudhary, A., 2024. A novel approach for environmental impact assessment of road construction projects in India. *Env. Impact Assess. Rev.* 106, 107477. <https://doi.org/10.1016/j.eiar.2024.107477>.
- Chen, Y., Pei, Z., Feng, D., Yi, J., 2025. Environmental impact and comprehensive value assessment of epoxy-based binder for asphalt pavement recycling based on LCA and LCC. *Chem. Eng. J.* 511, 161973. <https://doi.org/10.1016/j.cej.2025.161973>.
- China Building Standard Design & Research Institute, 2015. *Standard Atlas for Urban Road-Asphalt Pavement (15MR201)*. China Planning Press, Beijing.
- Chong, C.H., Zhou, X., Zhang, Y., Ma, L., Bhutta, M.S., Li, Z., Ni, W., 2023. LMDI decomposition of coal consumption in China based on the energy allocation diagram of coal flows: an update for 2005–2020 with improved sectoral resolutions. *Energy* 285, 129266. <https://doi.org/10.1016/j.energy.2023.129266>.
- Chu, J., Zhou, Y., Cai, Y., Wang, X., Li, C., Liu, Q., 2022. Flow and stock accumulation of plastics in China: patterns and drivers. *Sci. Total Environ.* 852, 158513. <https://doi.org/10.1016/j.scitotenv.2022.158513>.
- Dai, T., Qu, Z., Shi, F., 2022. Infrastructure stock in the process of urbanization in Beijing. *Alex. Eng. J.* 61 (4), 3277–3291. <https://doi.org/10.1016/j.aej.2021.08.060>.
- Dai, T., Yue, Z., 2023. The evolution and decoupling of in-use stocks in Beijing. *Ecol. Econ.* 203, 107606. <https://doi.org/10.1016/j.ecolecon.2022.107606>.
- De Pascale, B., Tataranni, P., Indacochea-Vega, I., Rodriguez-Hernandez, J., Lantieri, C., Bonoli, A., 2025. Enhancing road performance and sustainability: a study on recycled porous warm mix asphalt. *Sci. Total Environ.* 960, 178370. <https://doi.org/10.1016/j.scitotenv.2025.178370>.
- Deng, T., Fu, C., Zhang, Y., 2022. What is the connection of urban material stock and socioeconomic factors? A case study in Chinese cities. *Resour. Conserv. Recycl.* 185, 106494. <https://doi.org/10.1016/j.resconrec.2022.106494>.
- Gao, Z., Zhang, Q., Liu, B., Liu, J., Wang, G., Ni, R., Yang, K., 2024. The driving factors and mitigation strategy of CO₂ emissions from China's passenger vehicle sector towards carbon neutrality. *Energy* 294, 130830. <https://doi.org/10.1016/j.energy.2024.130830>.
- Grossegger, D., MacAskill, K., Al-Tabbaa, A., 2024. A critical review of road network material stocks and flows: current progress and what we can learn from it. *Resour. Conserv. Recycl.* 205, 107584. <https://doi.org/10.1016/j.resconrec.2024.107584>.
- Gu, S., Fu, B., Thriveni, T., Fujita, T., Ahn, J.W., 2019. Coupled LMDI and system dynamics model for estimating urban CO₂ emission mitigation potential in Shanghai, China. *J. Clean. Prod.* 240, 118034. <https://doi.org/10.1016/j.jclepro.2019.118034>.
- Han, J., Chen, W.-Q., Zhang, L., Liu, G., 2018. Uncovering the spatiotemporal dynamics of urban infrastructure development: a high spatial resolution material stock and flow analysis. *Env. Sci. Technol.* 52 (21), 12122–12132. <https://doi.org/10.1021/acs.est.8b03111>.
- Hasan, M.M., Liu, K., 2022. Decomposition analysis of natural gas consumption in Bangladesh using an LMDI approach. *Energy Strategy Rev.* 40, 100724. <https://doi.org/10.1016/j.esr.2021.100724>.
- Hossain, M.D.U., Wong, J.J.Y., Ng, S.T., Wang, Y., 2022. Sustainable design of pavement systems in highly urbanized context: a lifecycle assessment. *J. Env. Manage* 305, 114410. <https://doi.org/10.1016/j.jenvman.2021.114410>.
- Hu, H., Vizzari, D., Zha, X., Mantalovas, K., 2023. A comparison of solar and conventional pavements via life cycle assessment. *Transp. Res. D: Transp. Environ.* 119, 103750. <https://doi.org/10.1016/j.trd.2023.103750>.
- Huang, Y., Wolfram, P., Miller, R., Azarijafari, H., Guo, F., An, K., Li, J., Hertwich, E., Gregory, J., Wang, C., 2022. Mitigating life cycle GHG emissions of roads to be built through 2030: case study of a Chinese province. *J. Env. Manage* 319, 115512. <https://doi.org/10.1016/j.jenvman.2022.115512>.
- Huijbregts, M.A.J., Steinmann, Z.J.N., Elshout, P.M.F., Stam, G., Veronesi, F., Vieira, M., Zijp, M., Hollander, A., Van Zelm, R., 2017. ReCiPe2016: a harmonised life cycle impact assessment method at midpoint and endpoint level. *Int. J. Life Cycle Assess.* 22 (2), 138–147. <https://doi.org/10.1007/s11367-016-1246-y>.
- Karjalainen, T., Karjalainen, V., Waga, K., Tokola, T., 2024. Predicting the roadway width of forest roads by means of airborne laser scanning. *Int. J. Appl. Earth Obs. Geoinf.* 133, 104109. <https://doi.org/10.1016/j.jag.2024.104109>.
- Kaya, Y., 1989. *Impact of Carbon Dioxide Emission Control On GNP growth: Interpretation of Proposed Scenarios*. Intergovernmental Panel on Climate Change/Response Strategies Working Group.
- Klooststra, B., Makarchuk, B., Saxe, S., 2022. Bottom-up estimation of material stocks and flows in Toronto's road network. *J. Ind. Ecol.* 26 (3), 875–890. <https://doi.org/10.1111/jiec.13229>.
- Li, Y., Wang, S., Chen, B., 2019. Driving force analysis of the consumption of water and energy in China based on LMDI method. *Energy Procedia* 158, 4318–4322. <https://doi.org/10.1016/j.egypro.2019.01.790>.
- Liu, T., Zhao, R., Xie, Z., Xiao, L., Chen, A., Feng, W., You, Z., Feng, M., Li, R., 2023. Carbon emissions from accumulated stock of building materials in China. *Build. Env.* 240, 110451. <https://doi.org/10.1016/j.buildenv.2023.110451>.
- Long, Y., Song, Q., Huang, B., Zeng, X., Wu, H., 2024. Characterizing temporal and spatial characteristics of urban building material metabolism and embodied carbon emissions through a 4D GIS-MFA-LCA model. *Resour. Conserv. Recycl.* 206, 107642. <https://doi.org/10.1016/j.resconrec.2024.107642>.
- Mao, R., Bao, Y., Huang, Z., Liu, Q., Liu, G., 2020. High-resolution mapping of the urban built environment stocks in Beijing. *Env. Sci. Technol.* 54 (9), 5345–5355. <https://doi.org/10.1021/acs.est.9b07229>.
- Meijer, A., Arora, M., Cheah, L., 2024. Material stock-service and circularity prospects of buildings in Singapore. *Resour. Conserv. Recycl.* 208, 107697. <https://doi.org/10.1016/j.resconrec.2024.107697>.
- Miatto, A., Dawson, D., Nguyen, P.D., Kanaoka, K.S., Tanikawa, H., 2021. The urbanisation-environment conflict: insights from material stock and productivity of transport infrastructure in Hanoi, Vietnam. *J. Env. Manage* 294, 113007. <https://doi.org/10.1016/j.jenvman.2021.113007>.
- Ministry of Housing and Urban-Rural Development of the People's Republic of China, 2016. *Code For Design of Urban Road Engineering (CJJ 37-2012, 2016 Edition)*. China Architecture & Building Press.
- Mollaei, A., Ibrahim, N., Habib, K., 2021. Estimating the construction material stocks in two Canadian cities: a case study of Kitchener and Waterloo. *J. Clean. Prod.* 280, 124501. <https://doi.org/10.1016/j.jclepro.2020.124501>.
- Nguyen, T.C., Miatto, A., Fishman, T., Kim, J., 2023. The stock-service productivity of the European road transport infrastructure. *Resour. Conserv. Recycl.* 193, 106961. <https://doi.org/10.1016/j.resconrec.2023.106961>.
- Ortega-Ruiz, G., Mena-Nieto, A., García-Ramos, J.E., 2020. Is India on the right pathway to reduce CO₂ emissions? Decomposing an enlarged Kaya identity using the LMDI method for the period 1990–2016. *Sci. Total Environ.* 737, 139638. <https://doi.org/10.1016/j.scitotenv.2020.139638>.
- Qiao, Y., Wang, Z., Meng, F., Parry, T., Cullen, J., Liu, S., 2022. Evaluating the economic and environmental impacts of road pavement using an integrated local sensitivity model. *J. Clean. Prod.* 371, 133615. <https://doi.org/10.1016/j.jclepro.2022.133615>.
- Rousseau, L.S.A., Klooststra, B., Azarijafari, H., Saxe, S., Gregory, J., Hertwich, E.G., 2022. Material stock and embodied greenhouse gas emissions of global and urban road pavement. *Environ. Sci. Technol.* 56 (24), 18050–18059. <https://doi.org/10.1021/acs.est.2c05255>.
- Tareq, M.F., Deetman, S., Tukker, A., 2025. Evaluating the development of macro infrastructural stocks in the Netherlands: assessing factors driving historical material stocks growth. *Resour. Conserv. Recycl. Adv.* 27, 200269. <https://doi.org/10.1016/j.rcradv.2025.200269>.
- Wang, F., Zhang, Z., 2022. Decoupling economic growth from energy consumption in top five energy consumer economies: a technological and urbanization perspective. *J. Clean. Prod.* 357, 131890. <https://doi.org/10.1016/j.jclepro.2022.131890>.

- Wang, S., Zhou, Q., Tian, Y., 2020. Understanding completeness and diversity patterns of OSM-based land-use and land-cover dataset in China. *ISPRS Int. J. Geoinf.* 9 (9), 531. <https://doi.org/10.3390/ijgi9090531>.
- Wang, Z., Wiedenhofer, D., Stephan, A., Perrotti, D., Van Den Bergh, W., Cao, Z., 2023. High-resolution mapping of material stocks in Belgian road infrastructure: material efficiency patterns, material recycling potentials, and greenhouse gas emissions reduction opportunities. *Env. Sci. Technol.* 57 (34), 12674–12688. <https://doi.org/10.1021/acs.est.2c08703>.
- Wiedenhofer, D., Baumgart, A., Matej, S., Virág, D., Kalt, G., Lanau, M., Tingley, D.D., Liu, Z., Guo, J., Tanikawa, H., Haberl, H., 2024b. Mapping and modelling global mobility infrastructure stocks, material flows and their embodied greenhouse gas emissions. *J. Clean. Prod.* 434, 139742. <https://doi.org/10.1016/j.jclepro.2023.139742>.
- Wiedenhofer, D., Schug, F., Gauch, H., Lanau, M., Drewniok, M.P., Baumgart, A., Virág, D., Watt, H., Serrenho, A.C., Tingley, D.D., Haberl, H., Frantz, D., 2024a. Mapping material stocks of buildings and mobility infrastructure in the United Kingdom and the Republic of Ireland. *Resour. Conserv. Recycl.* 206, 107630. <https://doi.org/10.1016/j.resconrec.2024.107630>.
- Wolman, A., 1965. The metabolism of cities. *Sci. Am.* 213 (3), 178–190.
- Xu, Y., Zhang, H., Weng, D., Wen, X., Wang, S., Xie, Z., Hu, X., Zhang, L., 2025. An integrated assessment of road transportation carbon emissions in a rapidly urbanizing megaregion: evidence from the KUA-GBA, China. *J. Clean. Prod.* 525, 146417. <https://doi.org/10.1016/j.jclepro.2025.146417>.
- Yang, D., Xiao, B., Lu, X., Jia, X., Li, X., Han, F., Sun, L., Shi, F., Khumvongsa, K., Li, J., Duan, X., 2023a. Assessment and driving factor of housing vacancies in Shandong Peninsula urban agglomeration based on multi-source remote sensing data. *Heliyon* 9 (6), e16837. <https://doi.org/10.1016/j.heliyon.2023.e16837>.
- Yang, X., Hu, M., Shan, W., Zhang, C., Li, T., Pan, Y., 2023b. Integrating bottom-up building stock model with logistics networks to support the site selection of circular construction hub. *J. Clean. Prod.* 430, 139677. <https://doi.org/10.1016/j.jclepro.2023.139677>.
- Ye, L., Tan, S.-Y., Telesford, J.N., 2026. 3D GIS modelling of road and building material stocks: a case study of Grenada. *Resour. Conserv. Recycl.* 225, 108605. <https://doi.org/10.1016/j.resconrec.2025.108605>.
- Yu, B., Chen, Q., Li, N., Wang, Y., Li, L., Cai, M., Zhang, W., Gu, T., Zhu, R., Zeng, H., Zhang, Y., Jiang, J., 2024. Life cycle assessment of urban road networks: quantifying carbon footprints and forecasting future material stocks. *Constr. Build. Mater.* 428, 136280. <https://doi.org/10.1016/j.conbuildmat.2024.136280>.
- Yu, B., Li, L., Tian, X., Yu, Q., Liu, J., Wang, Q., 2021. Material stock quantification and environmental impact analysis of urban road systems. *Transp. Res. D: Transp. Environ.* 93, 102756. <https://doi.org/10.1016/j.trd.2021.102756>.
- Zhang, R., Yamashita, N., Liu, Z., Guo, J., Hiruta, Y., Shirakawa, H., Tanikawa, H., 2023. Paving the way to the future: mapping historical patterns and future trends of road material stock in Japan. *Sci. Total Environ.* 903, 166632. <https://doi.org/10.1016/j.scitotenv.2023.166632>.
- Zhang, T., Liu, L., Lv, X., 2019. The change in the material stock of urban infrastructures in China. *Struct. Change Econ. Dyn.* 51, 24–34. <https://doi.org/10.1016/j.strueco.2019.06.014>.
- Zhao, F., Wu, H., Zhu, S., Zeng, H., Zhao, Z., Yang, X., Zhang, S., 2023. Material stock analysis of urban road from nighttime light data based on a bottom-up approach. *Env. Res.* 228, 115902. <https://doi.org/10.1016/j.envres.2023.115902>.
- Zhao, G., Zheng, X., Yuan, Z., Zhang, L., 2017. Spatial and temporal characteristics of road networks and urban expansion. *Land (Basel)* 6 (2), 30. <https://doi.org/10.3390/land6020030>.
- Zhao, X., Jiang, Y., Zhao, J., Wang, Y., Cai, Z., Zeng, M., Dong, J., Cong, W., He, K., Liu, H., 2025. Carbon emission inventory and mitigation potential for private cars in Chinese cities. *Transp. Res. D: Transp. Environ.* 140, 104614. <https://doi.org/10.1016/j.trd.2025.104614>.
- Zhong, L., Liu, X., Yang, P., Zhong, X., Zeng, X., Zou, C., Xu, X., 2022. Quantifying the spatiotemporal evolution of the in-use steel stock in countries along the Belt and Road. *Resour. Conserv. Recycl.* 181, 106226. <https://doi.org/10.1016/j.resconrec.2022.106226>.